



Faculty of Engineering & Technology

Computer Science Department

Software Engineering

COMP433

Phase Two: Requirements Engineering

Elicitation & Discovery

Business: Vet Doctor – Veterinary Home Visit Service

Prepared by: Developer Group 10

Instructor: Prof. Adel Taweel

Section: 3

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Team Members

Name	ID	Role
Noor Nasir	1220769	Project Manager
Wafaa Shaheen	1221769	QA / Software Tester
Ellen Izz	1220806	Requirement Engineer
Yousef Qawwas	1220137	Technical Architect
Fadi Bassous	1221005	Programmer

Load Distribution

During Phase 2 execution, all team members collaborated effectively, leveraging teamwork and communication skills to produce coherent and consistent tasks. The project manager, Noor, assigned each member responsibilities that best matched their strengths. Ellen, Yousef, and herself were appointed as leads for the User Requirements, System Requirements, and Effort and Time Estimation tasks, respectively. This distribution of responsibilities improved the efficiency of the collaborative work process, while all team members continued to contribute significantly to each task through active participation and support.

Meeting Minutes

Meeting No.	Date & Time	Location	Attendees	Topics Discussed	Outcomes
1	25/04/2026, 6:00 PM	Online	Members of both groups	Discussed business overview, service types (Routine Check-Up, Emergency Visit, Farm Visit), user types (pet owners, farmers, veterinarians), and booking process	Confirmed three service types and main system actors. Agreed on business scope and services to be supported by the system
2	3/5/2026, 7:00 PM	Online	Members of both groups	Reviewed drafted user and system requirements UR1-UR9, discussed emergency visit priority handling, payment methods, admin responsibilities, and veterinarian approval process	Agreed on final user requirements and system requirement details. Confirmed payment options and emergency handling approach

User Requirements

Lead: Ellen Izz

Contributors: [Noor Nasir – reviewing], [Yousef Qawwas – discussion], [Wafaa Shaheen – finalising]

UR1.0: The system shall allow **clients**, including pet owners and farmers, and **veterinarians** to **register new accounts**.

UR2.0: The system shall allow registered **clients** and **veterinarians** to **log in** securely using their credentials.

UR3.0: The system shall enable **clients** (pet owners and farmers) to **book veterinary home or farm visits** by **specifying the service type, animal details, preferred date and time, and visit location**.

UR4.0: The system shall allow registered **clients** to **browse veterinarian profiles, including specializations, service areas, availability calendars, ratings, reviews, and aggregated feedback**.

UR5.0: The system shall enable **veterinarians** to **manage appointment-related information**, including **schedules, visit statuses, post-visit medical records, and prescriptions**.

UR6.0: The system shall enable registered **clients** to make payments securely using multiple payment methods.

UR7.0: The system shall enable clients to track the status of their booked appointments and communications with their assigned veterinarian.

UR8.0: The administrator shall be able to manage user accounts and monitor overall business performance.

UR9.0: The system shall allow registered clients to submit feedback for completed veterinary visits, including ratings and written reviews.

UR10.0: The system shall provide clients with informational content including service guides, how-to booking instructions, and general information about the veterinary service.

System Requirements

Lead: Yousef Qawwas

Contributors: [Ellen Izz – reviewing], [Fadi Bassous – discussion], [Wafaa Shaheen – finalising]

UR1.0: The system shall allow clients, including pet owners and farmers, and veterinarians to register new accounts.

- **SR1.1:** The system shall provide separate registration forms for clients and veterinarians. The client registration form shall collect the user's full name, phone number, email address, physical address, account type, and password. The veterinarian registration form shall additionally collect the professional license number, specialization, and service area.
- **SR1.2:** The system shall validate that the provided email address follows a valid format, such as user@domain.com, and verify that the email address is unique before completing registration. If the provided email address already exists in the database, the system shall reject the registration and notify the user.
- **SR1.3:** The system shall require passwords to be at least 8 characters long and contain at least one uppercase letter, one digit, and one special character. Passwords shall be stored using bcrypt hashing.
- **SR1.4:** The system shall provide a pre-configured administrator account managed exclusively by the system owner and shall prevent administrator accounts from being created through the public registration form.

UR2.0: The system shall allow registered clients and veterinarians to log in securely using their credentials.

- **SR2.1:** The system shall authenticate users by verifying their email address and password during login. If the provided email address or password is incorrect, the system shall reject the login attempt.
- **SR2.2:** The system shall temporarily lock the user account for 15 minutes after 5 consecutive failed login attempts and shall notify the user by email when the account is locked.
- **SR2.3:** The system shall provide a password reset option for registered users by sending a secure time-limited reset link to the user's registered email address.

- **SR2.4:** The system shall make the password reset link valid for 30 minutes. If the reset link expires before use, the system shall require the user to submit a new password reset request.

UR3.0: The system shall enable clients (pet owners and farmers) to book veterinary home or farm visits by specifying the service type, animal details, preferred date and time, and visit location.

- **SR3.1:** The system shall allow a logged-in client to initiate a booking by selecting a service type from the following options: Routine Check-Up, Emergency Visit, or Farm Visit. The client shall also specify the animal type (e.g., dog, cat, sheep, cow) and provide a brief description of the reason for the visit.
- **SR3.2:** The system shall require the client to confirm their visit location, either auto-populated from their registered address or manually entered, and select a preferred appointment date and time from a calendar displaying only currently available slots.
- **SR3.3:** The system shall provide the client the option to either select a specific veterinarian from a filtered list based on specialization and availability, or opt for automatic assignment, in which the system assigns the nearest available qualified veterinarian based on the client's location and selected service type.
- **SR3.4:** If the preferred time slot is unavailable, the system shall automatically suggest the next three available time slots and allow the client to select one.
- **SR3.5:** The system shall flag Emergency Visit bookings as high priority and immediately notify the assigned veterinarian, who shall be required to acknowledge the booking within 15 minutes.
- **SR3.6:** The system shall allow clients to cancel or reschedule a booked appointment, and shall notify both the client and the assigned veterinarian of any changes made. Cancellations made less than 2 hours before the scheduled visit time shall trigger a cancellation fee notification to the client, as defined by the administrator-configured policy.

UR4.0: The system shall allow registered clients to browse veterinarian profiles, view their specializations, service areas, availability calendars, ratings, and reviews.

- **SR4.1:** The system shall display a browsable list of registered veterinarians, each showing their full name, professional photo, years of experience, specialization area (small animals, livestock, or both), and availability status (Available / Unavailable) based on their calendar for the current day.
- **SR4.2:** Each veterinarian profile shall display an aggregated rating (out of 5 stars) calculated from all verified client reviews, where a verified review is defined as one submitted by a client with a completed appointment with that veterinarian, along with the most recent 5 reviews with dates.
- **SR4.3:** The system shall provide a filter and search function allowing clients to narrow veterinarian listings by specialization, availability on a specific date, minimum rating, and proximity to the client's registered address or a manually entered location. If no veterinarians match the applied filters, the system shall display a clear message and suggest broadening the search criteria.
- **SR4.4:** The system shall display the veterinarian's service area on a map, showing the geographic boundaries within which the veterinarian is available for home visits. The map shall be updated automatically whenever the veterinarian modifies their service area in their profile.
- **SR4.5:** The system shall display an interactive availability calendar on each veterinarian's profile page, showing available and booked time slots for the next 30 days. Time slots already booked shall be marked as unavailable and shall not be selectable during the booking process.

UR5.0: The system shall enable registered veterinarians to manage their availability calendars, update the status of their assigned appointments in real time, and submit post-visit medical records and prescriptions following each completed visit.

- **SR5.1:** The system shall provide each logged-in veterinarian with a personalized dashboard displaying their upcoming appointments for the day, week, or month sorted chronologically, with service type displayed as a label on each entry.
- **SR5.2:** The system shall allow the veterinarian to update the status of each appointment through defined stages: Confirmed, En Route, In Progress, and Completed. The system shall automatically notify the client via the platform whenever the status of their appointment changes.

- **SR5.3:** Upon marking a visit as Completed, the system shall prompt the veterinarian to upload visit notes (free text, max 2,000 characters), a digital prescription (PDF or structured form), and any relevant files (photos, lab results; max file size 10 MB per file). These records shall be securely stored and attached to the client's animal profile, and shall be accessible to the client through their account.
- **SR5.4:** The system shall allow veterinarians to flag a completed appointment for follow-up, specifying a follow-up date. On the specified date, the system shall automatically send a reminder notification to both the veterinarian and the client via email and in-app notification.
- **SR5.5:** The system shall allow veterinarians to set and update their availability calendar, marking specific days or time slots as unavailable. Changes shall be reflected in the booking system in real time, preventing new bookings for those slots. If a time slot marked unavailable already has a confirmed booking, the system shall alert the veterinarian and require them to manually cancel or reschedule the affected appointment before the block takes effect.
- **SR5.6:** The system shall allow veterinarians to view the full medical history and previous visit records of a client's animal before or during an appointment, including past prescriptions, diagnoses, and uploaded files. Access shall be restricted to records associated with animals linked to the veterinarian's own appointments.

UR6.0: The system shall enable registered clients to make payments securely using multiple payment methods and receive digital invoices upon service completion.

- **SR6.1:** The system shall present clients with the following payment options at the time of booking or upon service completion: Cash on Delivery, Credit/Debit Card (via a PCI-DSS compliant payment gateway), or a supported E-wallet service. The selected payment method shall be recorded and associated with the booking.
- **SR6.2:** The system shall automatically calculate the total invoice amount based on the base service price for the selected service type, plus any additional charges for medications, laboratory tests, or extended service time, as entered by the veterinarian post-visit. Additional charges entered by the veterinarian shall require client acknowledgment before the final invoice is generated.
- **SR6.3:** Upon successful online payment, the system shall immediately generate and send a digital invoice to the client's registered email address and display it within their account. The invoice shall include the service date, veterinarian name, service type, itemized charges, and total amount paid. For Cash on Delivery payments, the system shall generate

and display the invoice within the client's account upon the veterinarian marking the visit as Completed and confirming cash receipt.

- **SR6.4:** The system shall maintain a secure transaction history for each client and veterinarian, accessible from their respective dashboards, with the ability to filter by date range and download past invoices as PDF files.
- **SR6.5:** The system shall not store, log, or transmit client card details in plain text. All payment data shall be handled exclusively through the PCI-DSS compliant gateway, and the system shall only retain a masked reference (e.g., last 4 digits) for display purposes in transaction history.

UR7.0: The customer shall be able to track the status of their booked appointments and communicate with their assigned veterinarian before and during the service.

- **SR7.1:** The system shall automatically send a booking confirmation notification to both the customer and the assigned veterinarian within 30 seconds of a successful appointment being created, including appointment date, time, location, and service type.
- **SR7.2:** The system shall display the real-time status of each appointment to the customer, with the following defined stages: Booking Confirmed, Veterinarian Assigned, Veterinarian On the Way, Visit in Progress, and Visit Completed.
- **SR7.3:** The system shall send automated reminder notifications to the client 24 hours and 1 hour before the scheduled appointment.
- **SR7.4:** The system shall provide an in-app messaging channel between a customer and their assigned veterinarian, accessible only for appointments that are in an active or confirmed state, allowing both parties to exchange text messages and image attachments.
- **SR7.5:** The system shall retain and display the full communication history for each appointment for a minimum period of 12 months, accessible to both the customer and the veterinarian involved in that appointment.
- **SR7.6:** The system shall enable customers to request an online consultation session with their assigned veterinarian through the appointment details page, supporting voice or video communication.

- **SR7.7:** The system shall allow the assigned veterinarian to accept or decline an online consultation request; upon acceptance, the system shall automatically initiate the voice or video session between both parties.

UR8.0: The administrator shall be able to manage user accounts, oversee system operations, and monitor overall business performance through a dedicated administration interface.

- **SR8.1:** The system shall provide the administrator with a central management dashboard displaying real-time statistics, including total active clients, registered veterinarians, bookings scheduled for the current day, pending payments, and active system alerts.
- **SR8.2:** The system shall require all new veterinarian registrations to be reviewed and explicitly approved by the administrator before the account is activated, including verification of uploaded professional credentials such as license number and degree certificate.
- **SR8.3:** The system shall enable the administrator to deactivate, suspend, or permanently delete any user account, whether a client or a veterinarian, and shall automatically log the action, the stated reason, and the timestamp for audit purposes.
- **SR8.4:** The system shall automatically generate a monthly performance report accessible to the administrator, detailing total bookings by service type, total revenue and profit margin, individual veterinarian revenue contributions, and client acquisition and retention rates. The report shall be exportable as a PDF or spreadsheet file.
- **SR8.5:** The system shall enable the administrator to configure and update service pricing for each service type, notification message templates, and supported payment methods.

UR9.0: The system shall enable registered clients to submit ratings and written reviews for completed veterinary visits, and shall display aggregated feedback on veterinarian profiles.

- **SR9.1:** The system shall automatically prompt the client to submit a rating and review within 24 hours of a visit being marked as Completed. If the client does not respond within 72 hours of the prompt, the review request shall expire and the client shall no longer be able to submit a review for that appointment.
- **SR9.2:** The rating shall be submitted on a 1-to-5 star scale, with an optional free-text review field of up to 500 characters. The system shall permit only one review submission per

completed appointment, and shall prevent review submission for appointments that were cancelled or incomplete

- **SR9.3:** Submitted reviews shall be associated with the specific appointment and displayed on the veterinarian's public profile within 1 hour of submission after passing an automated content moderation check for inappropriate language. Reviews flagged by the automated check shall be held for manual administrators review before publication.
- **SR9.4:** The system shall calculate and display each veterinarian's average rating dynamically, updating immediately upon each new review submission.
- **SR9.5:** The system administrator shall be able to view all submitted reviews and permanently remove any review that contains inappropriate, offensive, or false content. The removal action shall be logged with a timestamp and reason for audit purposes. The veterinarian's average rating shall be recalculated automatically upon removal.

UR10.0: The system shall provide clients with informational content including service guides, how-to booking instructions, and general information about the veterinary service.

- **SR10.1:** The system shall provide a dedicated informational section accessible to all visitors, including non-registered users, containing a description of the business, the types of services offered, and the areas served.
- **SR10.2:** The system shall provide a step-by-step how-to booking guide explaining the booking process to new clients, including how to register, select a service type, choose a veterinarian, and confirm an appointment.
- **SR10.3:** The system shall provide a frequently asked questions section covering common client inquiries such as payment methods, cancellation procedures, emergency visit handling, and veterinarian qualifications.

Effort and Time Estimation

Lead: Noor Nasir

Contributors: [Wafaa Shaheen – reviewing], [Yousef Qawwas – discussion], [Fadi Bassous – finalising]

User Requirement	Estimated No. of Developers	Estimated Effort Per Developer	Total Effort
UR1: Account Registration	1	2 pw	$1 \times 2 = 2$ pw
UR2: System Login	1	1 pw	$1 \times 1 = 1$ pw
UR3: Appointment Booking	2	3 pw	$2 \times 3 = 6$ pw
UR4: Veterinarian Profile Browsing	1	2 pw	$1 \times 2 = 2$ pw
UR5: Veterinarian Schedule and Visit Management	2	3 pw	$2 \times 3 = 6$ pw
UR6: Payment Processing	2	2 pw	$2 \times 2 = 4$ pw
UR7: Real-Time Tracking and Communication	2	3 pw	$2 \times 3 = 6$ pw
UR8: Administrator Dashboard and Reporting	2	2 pw	$2 \times 2 = 4$ pw
UR9: Customer Feedback and Rating	1	2 pw	$1 \times 2 = 2$ pw
UR10: Informational Content and Service Guides	1	1 pw	$1 \times 1 = 1$ pw
Total Effort / Avg	$=(1+1+2+1+2+2+2+2+1+1)/10$ $= 15/10 = 1.5$ devs	21 pw	34 pw
Schedule Time (30% overhead)	Min: $21 \times 1.30 = 28$ w Max: $34 \times 1.30 = 45$ w		
Cost	(avg salary \$250 / week) $250 \times 45 = \\$11,250$		
Profit Margin (min 10%, max 30%)	Min: $\$11,250 \times 1.10 = \$12,375$ Max: $\$11,250 \times 1.30 = \$14,625$		

Note: pw = person-week. Schedule time is in calendar weeks including working days only. Average developer salary estimated at \$250/week. Final project price to the client ranges from **\$12,375 (min) to \$14,625 (max)**